# Computational Study

Domain Specialist

Inverse Methods Expert

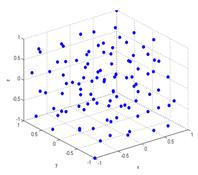
Research Software Engineer

Data Scientist

Parameter Range

Sample Size

Sampler



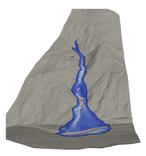
Geometry

Initial Condition

Boundary Condition

Model Parameter

Forward Model/ Simulator

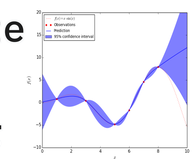


Design

Response

Inputs

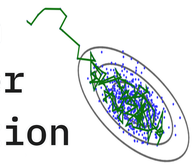
Surrogate Model/ Emulator



Outputs

Observation

Bayesian Parameter Calibration

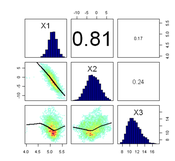


Calibrated Parameter

Performance Indicator

Statistics

Diagnosis




Input Data

**Computational Unit**

**Model/ Algorithm**

>>
10101010101
00100101000
01011000

**Compute Environment**

Software Ecosystem

Hardware Platform

Output Data

## Inputs

Geometry

BC/IC

Parameter

Sample

## Outputs

Field

QoI

Visual

Report

**Data**

OpenFOAM

RobustGaSP

SALIB

emcee The MCMC Hammer

ArviZ

**Models & Algorithms**

## Hardware Platform

CPU

GPU

Local

HPC

Cloud

## Software Ecosystem

**Compute Environment**